

Dr Shamendra Egodawela

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Professional Summary

Teaching and research academic in AI, machine learning, and intelligent systems, with experience in undergraduate and postgraduate teaching, student supervision, and industry-engaged research. Currently a Postdoctoral Research Fellow at the University of Technology Sydney (UTS), applying AI and data-driven methods to large-scale engineering systems. Research spans computer vision, multi-sensor and embedded systems, and systems for AI, with a Q1 publication record and active industry collaboration.

Education

2021 – 2025	RMIT University, Melbourne, Australia <i>Doctor of Philosophy in Engineering</i>
2015 – 2020	University of Peradeniya, Sri Lanka <i>Bachelor of Science of Engineering (Electrical & Electronic Engineering)</i> CGPA: 3.65 / 4.00

Academic Experience

May 2025 – Present	University of Technology Sydney <i>Postdoctoral Research Fellow</i> - Designed and delivered postgraduate lectures in Control of Mechatronic Systems (Course ID: 49329), including development of course material, assessment activities, and project-based learning. - Leading research in an industry-partnered program between Amplitel, Australia’s largest telecommunications infrastructure provider, and UTS, applying AI and machine learning to asset condition assessment. - Developed AI-driven predictive models over large-scale structured and unstructured engineering data (maintenance records, geospatial and risk indicators) to identify failure modes and support proactive decision-making. - Built and evaluated machine learning and sensing pipelines for structural condition assessment and remote monitoring of critical infrastructure. - Supervised and coordinated 10+ research interns, providing technical guidance and mentorship in support of project delivery. - Collaborated with research and industry stakeholders, presenting results to support data-driven decision-making.
2023 – 2025	RMIT University <i>Casual Researcher & Course Assistant</i> - Delivered undergraduate teaching through lectures and laboratories in Mechanical, Manufacturing and Mechatronics Engineering (MMM), supporting hands-on and project-based learning. - Supervised student-led engineering projects and led development of a smart sensing platform integrating RFID, weighing systems, and machine learning for behaviour monitoring. - Developed an augmented reality-based laboratory model to support teaching of finite element concepts.

University of Peradeniya*Teaching Instructor*

- Delivered lectures, tutorials, and laboratory sessions in electrical machines, power systems, and robotics.
- Designed and assessed laboratory coursework involving electrical and embedded systems.
- Supervised undergraduate capstone projects in MATLAB, ANSYS, SOLIDWORKS, AutoCAD, and Arduino-based platforms.

Teaching & Supervision

- Delivered undergraduate and postgraduate teaching through lectures, tutorials, and laboratory sessions across computing, AI-, and engineering-oriented subjects.
- Supervised undergraduate and intern projects involving AI, machine learning, sensing systems, and embedded platforms.
- Supported hands-on and project-based learning aligned with practical, real-world applications.
- Experience with digital learning platforms (Moodle, Canvas) for course delivery and assessment.

Research Output**Journal Publications**

2025

1. **Egodawela, S.**, Gostar, A. K., Samith Buddika, H. A. D., Harischandra, W. A. N. I., Dhammika, A. J. & Mahmoodian, M. Metal loss defect detection and depth estimation using multi-spectral image analysis of cooling excited steel specimen with corrosion. *Scientific Reports* 15, 23894. doi:10.1038/s41598-025-88528-w [SJR Q1]
2. Mahmoodian, M. & **Egodawela, S.**. Stochastic Finite Element Methods for Reliability Analysis of Corroded and Fatigued Steel Elements. *Available at SSRN*. doi:10.2139/ssrn.4975106

2024

3. **Egodawela, S.**, Gostar, A. K., Buddika, H. A. D. S., Harischandra, N., Abeykoon, D., White, P. A. & Mahmoodian, M. Multispectral imaging and spectral unmixing for characterization of relative phase composition of steel corrosion products. *IEEE Sensors Journal* 24, 37879–37893. doi:10.1109/JSEN.2024.3428472 [SJR Q1]
4. **Egodawela, S.**, Khodadadian Gostar, A., Buddika, H. A. D. S., Dammika, A. J., Harischandra, N., Navaratnam, S. & Mahmoodian, M. A deep learning approach for surface crack classification and segmentation in unmanned aerial vehicle assisted infrastructure inspections. *Sensors* 24, 1936. doi:10.3390/s24061936 [SJR Q1]

Conference Publications

2022

5. **Egodawela, S.**, Buddika, H. A. D. S., Harischandra, W. A. N. I., Mahmoodian, M. & Mathota, M. R. S. N. M. Conservation and management of ancient stupa using digital twins: a case study of UAV-based 3D photogrammetric digitization of Rankoth Wehera Stupa in *International Conference on Sustainable Built Environment*, 493–508. doi:10.1007/978-981-99-3471-3_35

2021

6. Perera, I., Senavirathna, S., Jayarathne, A., **Egodawela, S.**, Godaliyadda, R., Ekanayake, P., Wijayakulasooriya, J., Herath, V. & Sathyaprasad, S. Vehicle tracking based on an improved DeepSORT algorithm and the YOLOv4 framework in *2021 10th International Conference on Information and Automation for Sustainability (ICIAfS)*, 305–309. doi:10.1109/ICIAfS52090.2021.9606052

7. **Egodawela, S.**, Herath, H. M. D. M. B., Willamuna, S. M. A. B., Ekanayake, M. P. B., Godaliyadda, G. M. R. I., Wijayakulasooriya, J. V., Herath, V. R. & Sathyaprasad, I. M. S. Vehicle detection and localization for autonomous traffic monitoring systems in unstructured crowded scenes in *2020 IEEE 15th International Conference on Industrial and Information Systems (ICIIS)*, 192–197. doi:10.1109/ICIIS51140.2020.9342663
8. **Egodawela, S.**, Herath, H. M. D. M. B., Ranaweera, R. D. & Wijayakulasooriya, J. V. Device to remotely track and locate the position of a child for safety in *2020 IEEE Integrated STEM Education Conference (ISEC)*, 1–7. doi:10.1109/ISEC49744.2020.9397849

Research Program

AI and Machine Learning for Intelligent Sensing and Systems

Developing data-driven and AI-enabled systems for diagnostics, condition assessment, and decision support over large-scale engineering data.

- Systems for AI: machine learning and computer vision pipelines for real-world inspection and diagnostics.
- Multi-sensor and embedded systems integration for real-time monitoring.
- Predictive modelling and risk-based frameworks over structured and unstructured data.
- Robotics-assisted inspection for remote infrastructure monitoring.

Industry Engagement

- Led industry collaboration and stakeholder engagement through the UTS–Amplitel project.
- Applied AI-driven modelling and large-scale engineering datasets to deliver real-world monitoring solutions.

Technical Skills

Programming	Python, C/C++, Arduino
ML / Data	PyTorch/TensorFlow workflows, computer vision, predictive modelling
Engineering Tools	MATLAB, ANSYS, SOLIDWORKS, AutoCAD
Systems	Embedded Systems, PLC, Sensor Systems
Teaching Tools	Moodle, Canvas

Awards

CommsDay Edison Awards 2025 — Best Telco AI Initiative (Winner)

Project Lead, *Project SMaRTER: AI-Driven Tower Fingerprinting* (Amplitel & University of Technology Sydney). Led the development and delivery of an industry-scale AI and digital twin platform for telecom tower asset management, recognised nationally as Australia's leading telecommunications AI initiative.

Best Undergraduate Project Award — Institute of Engineers Sri Lanka (IESL)

Awarded for an outstanding final year engineering project.

Best Guest Tech Talk Award — National Institute of Business Management (NIBM), Innovation Centre

Research Interests

- AI and Machine Learning Systems
- Computer Vision and Intelligent Sensing
- Robotics and Mechatronic Systems
- Structural Health Monitoring and Asset Management

Referees

Prof. Raymond Owen

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